MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1R5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentagrid converter

Outline --- 5-2(EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	-f,g5	a	g2,g4	g1	-f,g5	g3	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2 and 4	Ec1	Ec3	Ik	Rg1
Unit:	Vdc	Vdc	Vdc	Vdc	Vdc	mAdc	Ohms
Maximum:	1.6	100	75	---	0	6.5	---
Minimum:	1.2	---	---	---	---	---	---
TEST CONDITIONS:	1.4	90	45	0	0	---	100,000

GENERAL:

Qualification - Not required

(C) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	1	Ec3 = -1 Vdc; Eg1 = 15 Vac	0.65	II	Ic3	0	-0.6	uAdc
1256	Electrode current (anode)	-	Eg1 = 15 Vac	0.65	II	Ib	0.34	0.94	mAdc
1256	Electrode current (cathode)	-	Eg1 = 15 Vac	0.65	II	Ik	1.53	3.45	mAdc
1326	Conversion transconductance (2)	-	Eg1 = 15 Vac; Ef = 1.1 Vdc	0.65	II	Sc	130	340	umhos
1237	Oscillator grid current	2	Ef = 1.1 Vdc; anode floating	0.65	II	Ic1	125	---	uAdc
(C) 1246	Audio frequency noise	3	Ecal (0 vu) = 1.2 Vac; Ecal (50 mw) = 200 mVac	1.0	II	---	---	---	---
(C) 1201	Short and discontinuity detection	-	40 G	0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Ec1 = -5 Vdc; Rp = 10,000 ohms; Rg1 = 0	---	---	Ep	---	500	mVac
1301	Filament current	-		---	---	If	44	56	mA
1326	Conversion transconductance (1)	-	Eg1 = 15 Vac	---	---	Sc	160	340	umhos
1331	Direct-interelectrode capacitance	-	Without shield Without shield Without shield	---	---	(Cg3-a1) (Cp-a1) (Cg3-p	5.6 5.2 ---	8.4 8.8 0.4	pf pf pf
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1121	Base strain	5		---	---	---	---	---	---
2126	Glass strain	-		4.0	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	4	Group A; Ef = 1.4 Vac or Vdc; Ec2 and 4 = 67.5 Vdc; Ec1 = -16.0 Vdc; Esig = 17.5 Vac	---	---	---	---	---	---
	Life-test end points (500 hours):								
1326	Conversion transconductance (1) and	-	Ef = 1.4 Vdc	---	---	Sc	125	---	umhos
1237	Oscillator grid current	-	Ef = 1.4 Vdc	---	---	Ic1	100	---	uAdc

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. With converter oscillator test set having Rgl = 50,000 ohms and with grid-to-filament resonant impedance adjusted to 9,500 ohms.
3. Ebb = 90 Vdc; Ecc2 = 40 Vdc; Ecc3 = Ecc1 = -10 Vdc; Rp = 0.2 meg; Rg2+4 = 1000 ohms; Rgl = 1.2 meg; Rg3 = 0.2 meg; Ecal Ovu = 1.2 Vac; Ecal 50 mw = 200 mVac.
4. Bias of No. 1 grid may be obtained from a dc source of self bias.
5. Sampling inspection shall be in accordance with MIL-E-1 paragraph "Base-strain test, miniature tube, sampling (method 1121)."

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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